

Technology News

Soon, a mobile phone malaria diagnostic sensor

A team of researchers at the University of Glasgow is developing a device that uses mobile-phone derived technology to detect and separate red blood cells infected with malaria parasites. If successful, devices based on the technology could be mass produced for rapid and accurate diagnosis of malaria.

The team is exploiting surface acoustic wave devices, which are electronic components commonly found in TVs, mobile phones and other electronic display devices, to exert selective forces on malaria-infected red blood cells to separate them from uninfected red cells. Different cells respond to surface acoustic waves in different ways depending on their physical properties including their elasticity and their shape.

Since malaria parasites cause red cells to alter their elasticity and their shape, they should respond differently to surface acoustic waves at particular frequencies. The team plans to produce a handheld device which can identify infected cells quickly and cleanly.

"Diagnosing malaria can be difficult and time-consuming procedure because few blood cells actually carry the parasite. It can take clinicians hours to secure a positive or negative result on a blood sample. Moreover, developing a reliable, portable diagnostic tool is not only important for the individual concerned but it is also important in preventing the spread of drug resistance in the parasites," said Jon Cooper, a team member.

LED-grid glasses for partially-sighted people

Scientists at the Oxford University have created a pair of electronic glasses to help partially-sighted people see again.



Malaria diagnostic sensor

Led by Dr Stephen Hicks, the team from Oxford's Nuffield Department of Clinical Neurosciences has developed a prototype version of the glasses. The device uses grids of LED lights placed in front of the eyes, to represent nearby objects to individuals who have lost almost all their vision.

"Most people who lose their sight do so through age or disease. A lot of people can still see contrast, even if they can't see their hand in front of their face," said Iain Wilson of the Nuffield Department.

Stereoscopic cameras attached to



Electronic glasses for partially-sighted people

the glasses' rims capture visual and depth information. A display of tiny lights embedded in the see-through lenses of the glasses feed back extra information about objects, people or obstacles in view. Software then processes this data by recognising objects of interest and representing them as simple patterns, with brighter lights signalling nearer objects.

Future versions of the technology could use different colours to indicate different objects. It will also use smaller integrated cameras to make the glasses more comfortable, while mobile phones are likely to be powerful enough to run the software, eliminating the need for a separate computer system.

If the user turns his head, a gyroscope will indicate which direction he is facing and a few seconds' worth of memory will mean the system doesn't have to recapture the data if he turns back quickly, speeding up the processing.

'Talking' cars soon to become reality

You must have seen talking cars only in cartoon films or read about them in fictions. Very soon such cars might